

Ore-grade thermal data analysis

State of Thermal Data Analysis for Ore Systems

Ore-grade thermal data analysis encompasses both thermal infrared (TIR) spectroscopic characterization for deposit grading and microwave-induced thermal imaging for automated physical ore sorting (Wu et al., 2026; John et al., 2015). These techniques decode mineral emissivity and thermodynamic response to evaluate mineral concentration, alteration intensity, and mineral processing performance (Qi et al., 2023; Ghosh et al., 2014).

Spectral Emission and Geochemical Estimation

Thermal infrared sensing detects the fundamental vibrational modes of anhydrous silicates, carbonates, and sulfates that lack distinctive shortwave infrared features (Bedell, 2017; Qi et al., 2023). Processing TIR data facilitates direct mineral abundance extraction across multiple observation platforms (Laukamp et al., 2021; Hecker et al., 2019).

Analytical Scale	Sensor / Method	Targeted Ore System & Minerals	Analytical Performance
Laboratory Mixtures	Hyperspectral TIH with CARS-PLSR (Qi et al., 2023)	Potassium salts (picromerite, KCl) (Qi et al., 2023)	Model validation achieved \$R_p^2\$ of 0.943–0.948 (Qi et al., 2023).
Drill Core Profiling	HyLogger-3 TIR/SWIR joint unmixing (jCLST) (Green et al., 2025)	Multi-mineral modal abundances (Green et al., 2025)	Modal unmixing errors typically range between ±7–15 vol% (Green et al., 2025).
Quarry Mine Face	Ground-based Telops LWIR camera (Bouanga-Tombet et al., 2018)	Carbonate and silicate quarry lithologies (Bouanga-Tombet et al., 2018)	Radiometric variations reached 13 K across lithologic units (Bouanga-Tombet et al., 2018).
Airborne Profiling	TASI sensor with combined NEM-RATIO-MMD (Cui et al., 2015)	Diopside/hedenbergite in gold skarns (Cui et al., 2015)	Skarn samples averaged 0.73 g/t Au (Cui et al., 2015).
Satellite Mapping	Multispectral ASTER TIR and AVIRIS fusion (Ni et al., 2020)	Porphyry Cu-Au and epithermal deposits (Noori et al., 2019; Ni et al., 2020)	Classification accuracy increased by 1% to 5% (Ni et al., 2020).

FIGURE 1 Performance and sensor platforms across laboratory, core scanning, field mine face, airborne, and satellite scales.

In core scanning and airborne platforms, extracting diagnostic spectral features—such as quartz reststrahlen troughs (8.6–9.3 μm) and plagioclase peak shifts—provides direct proxies for hydrothermal alteration intensity and mineral chemistry (Wu et al., 2025; Hecker et al., 2019).

Microwave-Induced Sorting and Beneficiation

Microwave heating combined with thermal infrared thermography differentiates valuable mineral phases from gangue matrices based on differential dielectric heating (John et al., 2015; Ghosh et al., 2014):

- Microwave heating of 10 mm particles categorized iron ores containing **52 to 67 wt.% Fe** into distinct high-, medium-, and low-grade fractions based on individual particle peak temperature profiles (Ghosh et al., 2014).
- Numerical models of porphyry copper ores demonstrate that at **0.35% Cu**, the surface thermal differential between ore and gangue reaches $\sim 1^\circ\text{C}$ (John et al., 2015).
- Extending the post-heating delay time up to two minutes substantially minimizes thermal variation stemming from mineral grain size and texture (John et al., 2015).
- Thermogravimetry coupled with mass spectrometry identifies carbonaceous kerogen fragments (m/z 15, 29, 43) to assess **preg-robbing behavior** in refractory sulfide gold ores (Aleksandrova et al., 2019).

Deposit-Scale Thermodynamic and Inertia Modeling

At the deposit footprint level, thermal inertia and subsurface thermal modeling identify concealed ore zones (Wu et al., 2026; Ramakrishnan et al., 2013). Satellite and field thermal inertia mapping at the Mamandur polymetallic prospect detected distinct thermal inertia contrasts ($2118\text{--}5474 \text{ J}\cdot\text{m}^{-2}\cdot\text{K}^{-1}\cdot\text{s}^{-1/2}$) over Pb-Cu-Zn sulfide bodies beneath soil cover (Ramakrishnan et al., 2013).

Furthermore, integrating satellite thermal bands with digital elevation models through the Thermal Generalization Method targets high-purity barite mineralization at depths of **60 to 300 m**, resolving steep vertical temperature gradients ($\nabla T > 0.5 \text{ K/m}$) produced by exothermic precipitation (Kamaunji et al., 2025). In orogenic and Carlin-type gold systems, organic matter geothermometry ($T_{\text{max}} > 150^\circ\text{C}$) and microthermometry ($123\text{--}173^\circ\text{C}$) delineate the regional paleo-thermal fluid pathways directly governing mineralization (Bedini, 2017; Pinet et al., 2023).

Collectively, thermal data analysis delivers critical quantitative vectors across exploration targeting, core logging, and rapid ore pre-concentration (Bedell, 2017; Wu et al., 2025; John et al., 2015).

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